OCR H446 / PAPER 1 / UNIT 1.3

Exchanging data

CHAPTER CONTENTS

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Full-paper guide retains further synoptic practice and reference appendices.

OCR H446 / PAPER 1 / UNIT 1.3

Exchanging data

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H446/01 / REVISION

Understand it. Model it. Try it. Check it.

Use this chapter to understand the theory, follow worked models and practise independently.

FOUR SPECIFICATION FAMILIES

1.3.1 COMPRESSION + SECURITY

Lossless/lossy, RLE, dictionary, encryption and hashing

1.3.2 DATABASES

Keys, 3NF, SQL, referential integrity and transactions

1.3.3 NETWORKS

TCP/IP, DNS, routing, switching and security

1.3.4 THE WEB

HTML, CSS, JavaScript, search and client/server

FOUR PASSES FOR EVERY TOPIC

01 UNDERSTAND

Build a precise conceptual picture.

02 MODEL

Follow a worked example or trace.

03 TRY

Answer original exam-style questions.

04 CHECK

Self-mark with separate indicative points.

Practise SQL, networks, encryption and web technologies.

Use the OCR Appendix 5d/5f pages for SQL, HTML, CSS and ER diagrams.

1.3 / SPECIFICATION CHECKLIST

Every 1.3 topic has a working model

Jump to any topic using the linked rows. Each cycle has an explanation, a worked model, two practice questions and separate mark points.

SPEC	LEARNING CYCLE	PAGES
1.3.1(a)	Lossy and lossless compression	3-6
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1.3.4(a)	HTML, CSS and JavaScript	71-74
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1.3.4(d)	Client-side and server-side processing	79-82

SPECIFICATION BOUNDARY

Aligned to OCR H446 1.3.1-1.3.4. PageRank is taught conceptually, not as a required coded algorithm; relational normalisation is through 3NF.

Checked against: OCR H446 specification v2.7 (2024), plus OCR subject-content clarification (2020), sections 1.3.1-1.3.4. Questions and

1.3.1(a) / BUILD THE MENTAL MODEL

Smaller file; choose what can be removed

Compression represents information using fewer bits. The method depends on whether every original bit must be recovered.

THREE PIECES OF THE IDEA

LOSSLESS

The reconstructed data is identical to the original: suitable for code, accounts and exact text.

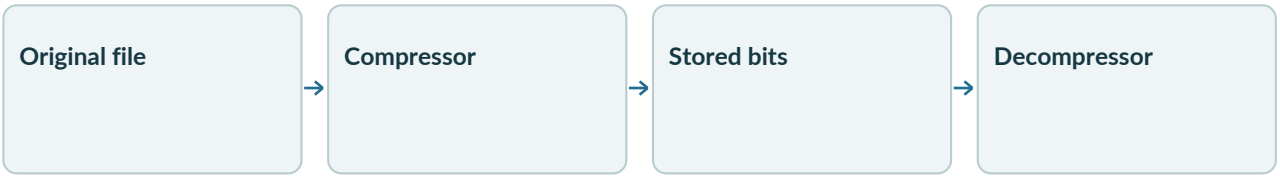
LOSSY

Discard selected information: suitable where acceptable perceptual quality matters more than exact recovery.

TRADE-OFF

Consider file size, quality, processing cost, use and whether later editing will occur.

MODEL / FOLLOW THE FLOW



Lossless returns the exact source. Lossy returns an approximation; repeated lossy saving can degrade quality.

EXAM-READY STATEMENT

Always justify the compression choice against the context; lossy is not automatically better because it is smaller.

WORDS THAT EARN MARKS

Lossless Identical output

Lossy Approximation

Ratio Original ÷ compressed

Artefact Visible or audible error

1.3.1(a) / STEP-BY-STEP WORKED EXAMPLE

Choose a method for three school files

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Source code	A program cannot tolerate missing characters	Use lossless compression so every character survives.
2. Photograph	Website thumbnail can trade some detail for size	A suitable lossy format may reduce transfer time.
3. Medical scan	Small features may matter diagnostically	Choose lossless if any data loss is unacceptable.
4. Calculate	12 MB compressed to 3 MB	Compression ratio = 12:3 = 4:1 (75% reduction).

TRANSFER IT TO AN EXAM ANSWER

Always justify the compression choice against the context; lossy is not automatically better because it is smaller.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

A school stores source code and photographs. Recommend a compression type for each and justify both.

QUESTION 2 | 4 MARKS

A file is reduced from 18 MB to 6 MB. Calculate the compression ratio and percentage reduction, and explain the result.

1.3.1(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Source code: lossless, because all characters/instructions must be exactly recovered.
- Photos: lossy may be appropriate if reduced size matters and small visual losses are acceptable.

Q2 / 4 MARKS

- Ratio = 18:6 = 3:1.
- Reduction = (18 - 6) / 18 × 100 = 66.7%.
- One third of the original size remains.

COMMON MISCONCEPTION

A compression ratio of 3:1 means the new file is one third the original, not that only 3% remains.

Green pen: State which property of the file makes lossless or lossy appropriate.

1.3.1(b) / BUILD THE MENTAL MODEL

Count the run; do not just delete repeats

Run-length encoding (RLE) replaces consecutive identical values with a count and the value. It is lossless if the representation can be decoded unambiguously.

THREE PIECES OF THE IDEA

ENCODE

Replace runs of a repeated symbol with a count and symbol.

DECODE

Reconstruct each run exactly using the stored count.

LIMITATION

Short alternating runs may create extra data rather than reduce it.

MODEL / FOLLOW THE FLOW

AAABBBBCC → 3A 4B 2C → Decoder → AAABBBBCC

In this teaching notation the encoded length is six tokens (3,A,4,B,2,C), compared with nine original characters.

EXAM-READY STATEMENT

RLE is effective when the source contains long runs of identical values, such as simple images with large solid areas.

WORDS THAT EARN MARKS

Run Repeated sequence

Count Run length

Lossless Exact recovery

Overhead Extra metadata

1.3.1(b) / STEP-BY-STEP WORKED EXAMPLE

Encode AAAAAABBBCC and compare an alternating string

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Identify runs	AAAAAA | BBB | CC	Runs have lengths 6, 3 and 2.
2. Encode	6A 3B 2C	Each stored pair gives count then symbol.
3. Decode	6A -> AAAAAA	All original information can be reconstructed.
4. Test failure	ABABAB -> 1A 1B 1A 1B 1A 1B	Encoding may be longer when repeats are scarce.
5. Consider format	Counts may need delimiters for multi-digit runs	Actual file size depends on how counts/symbols are represented.

TRANSFER IT TO AN EXAM ANSWER

RLE is effective when the source contains long runs of identical values, such as simple images with large solid areas.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Encode the symbols AAAABBBCCCAA using count-then-symbol RLE. Explain one reason the result may not always save space.

QUESTION 2 | 4 MARKS

A row of 12 white pixels is followed by one black pixel. Explain why RLE might work well for this image and how decoding restores it.

1.3.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Runs: 4A, 3B, 3C, 2A.
- Encoded as 4A3B3C2A in the stated notation.
- Every single-symbol run also needs a count, so alternating input can expand.

Q2 / 4 MARKS

- Store 12W 1B (with a known unambiguous format).
- Decoder outputs 12 white pixels then one black pixel.
- Long repeated runs yield fewer stored items than individual pixels.

COMMON MISCONCEPTION

RLE is not the same as removing duplicates anywhere in a file: only adjacent repeated symbols form a run.

Green pen: Check your result by decoding the sequence you created.

1.3.1(b) / BUILD THE MENTAL MODEL

One entry; reuse its shorter reference

Dictionary compression replaces recurring sequences with shorter references to entries in a shared dictionary.

THREE PIECES OF THE IDEA

BUILD

Identify repeating words, phrases or byte patterns and assign codes.

ENCODE

Write references to dictionary entries instead of repeating the full pattern.

DECODE

Use the same dictionary to rebuild the original data exactly.

MODEL / FOLLOW THE FLOW

Repeated phrases → **Dictionary entries** → **Index references** → **Exact original**

Simple illustrative example: the phrase COMPUTER repeats four times; a one-symbol token replaces later occurrences.

EXAM-READY STATEMENT

The coded message is not meaningful without an agreed dictionary; dictionary storage contributes to total compressed size.

WORDS THAT EARN MARKS

Dictionary Pattern-code map

Reference Short token

Lossless Exact recovery

Overhead Dictionary storage

1.3.1(b) / STEP-BY-STEP WORKED EXAMPLE

Substitute repeated colour names with a dictionary

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Source	RED BLUE RED BLUE RED	Repeated tokens are visible.
2. Dictionary	1 = RED; 2 = BLUE	Dictionary holds each full word once.
3. Encode	1 2 1 2 1	Every source token becomes a reference.
4. Decode	1 -> RED; 2 -> BLUE	Dictionary restores the exact token sequence.
5. Evaluate	Save size of codes minus dictionary cost	A large or infrequently reused dictionary may be inefficient.

TRANSFER IT TO AN EXAM ANSWER

The coded message is not meaningful without an agreed dictionary; dictionary storage contributes to total compressed size.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Create a two-entry dictionary for CAT DOG CAT DOG DOG and write the encoded message.

QUESTION 2 | 4 MARKS

Explain why a dictionary-based lossless method may perform poorly on a file where patterns seldom repeat.

1.3.1(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Example dictionary: A = CAT, B = DOG.
- Encoded data: A B A B B.
- Decoder uses each reference to restore the original order.

Q2 / 4 MARKS

- Few repeated patterns mean few substitutions.
- The dictionary itself takes storage space.
- References plus dictionary may be no smaller than the original.

COMMON MISCONCEPTION

A dictionary index is a compression reference, not a cryptographic secret or an encryption key.

Green pen: Explain how both sides access the dictionary and how this affects decompression.

1.3.1(c) / BUILD THE MENTAL MODEL

Which key is shared, and which stays private?

Encryption makes information unreadable to unauthorised parties. Symmetric and asymmetric schemes differ in their key use.

THREE PIECES OF THE IDEA

SYMMETRIC

The same shared secret key is used for encryption and decryption; generally efficient for bulk data.

ASYMMETRIC

Uses a public/private key pair; public encryption is decrypted with the matching private key.

HYBRID

Public-key methods can establish a shared session key; symmetric encryption then protects bulk traffic.

MODEL / FOLLOW THE FLOW

Public key step → **Session key** → **Symmetric cipher** → **Recipient decrypts**

A simplified secure connection often uses asymmetric techniques for key establishment/authentication and symmetric encryption for the data stream.

EXAM-READY STATEMENT

Avoid claiming that a recipient encrypts with their private key to keep a message confidential; that confuses confidentiality with signing.

WORDS THAT EARN MARKS

Plaintext	Readable data	**Ciphertext**	Encrypted data
Public key	Shared openly	**Private key**	Kept secret

1.3.1(c) / STEP-BY-STEP WORKED EXAMPLE

Alice sends a confidential file to Bob

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Obtain key	Alice obtains Bob's authentic public key	The identity of the key owner must be verified.
2. Protect	Alice encrypts a short secret for Bob	Only Bob's corresponding private key can recover it in this model.
3. Data	Shared secret protects the file symmetrically	Efficient bulk encryption uses the session key.
4. Receive	Bob recovers the secret and decrypts the file	Key secrecy protects confidentiality.
5. Check	Authentication is a separate concern	Encryption alone does not prove who sent a message.

TRANSFER IT TO AN EXAM ANSWER

Avoid claiming that a recipient encrypts with their private key to keep a message confidential; that confuses confidentiality with signing.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Compare symmetric and asymmetric encryption in terms of keys, speed and a realistic use case.

QUESTION 2 | 5 MARKS

Explain how a hybrid approach could protect a large file sent to a new recipient.

1.3.1(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Symmetric: same shared secret for encryption/decryption; efficient for bulk data.
- Asymmetric: public/private pair; suitable for securely establishing or transporting a session secret.
- Hybrid combines public-key key establishment with efficient symmetric data encryption.

Q2 / 5 MARKS

- Recipient has an authentic public key.
- Sender protects a session secret using a public-key mechanism.
- Recipient uses the corresponding private key to recover the secret.
- Both use the symmetric secret for bulk file encryption/decryption.

COMMON MISCONCEPTION

A public key can be public; the associated private key must not be sent to the other party.

Green pen: Label whose key is used at each stage and whether it is secret or public.

1.3.1(d) / BUILD THE MENTAL MODEL

A hash is a one-way fingerprint, not a cipher

A hash function maps input data to a fixed-size digest. Hashes help identify changes and support password verification with appropriate secure constructions.

THREE PIECES OF THE IDEA

DIGEST

The algorithm produces a fixed-length result from an arbitrary-length message.

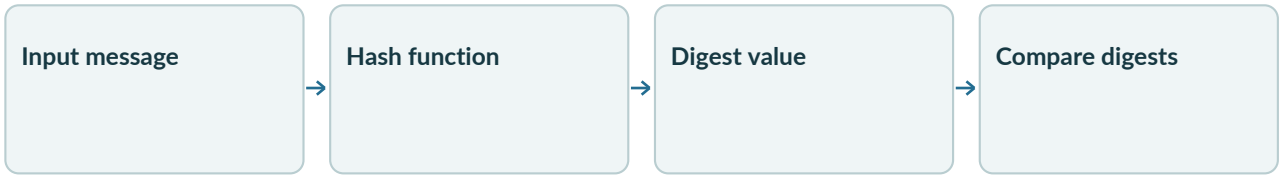
INTEGRITY

A changed digest suggests the data is different; accidental changes can be detected.

PASSWORDS

Store a salted, deliberately slow password-derived hash rather than the plaintext password.

MODEL / FOLLOW THE FLOW



A hash is recomputed from the received data. Matching digests support an integrity check but do not independently authenticate the source.

EXAM-READY STATEMENT

Hash collisions are theoretically possible. A plain fast hash alone is not a recommended modern password-storage design.

WORDS THAT EARN MARKS

Hash Fixed-length digest

Collision Same digest

Salt Unique random input

One-way Hard to invert

1.3.1(d) / STEP-BY-STEP WORKED EXAMPLE

Check a file and store a password safely

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. File digest	Record an expected digest	A receiver calculates a digest of the downloaded file.
2. Compare	Received digest differs	This indicates a mismatch; redownload or investigate.
3. Password set	Generate a unique random salt	Use a suitable password hashing/KDF scheme to store a derived value.
4. Password login	Recompute using stored salt	Compare the derived result in constant-time code.
5. Remember	Digest is not reversible encryption	There is no decryption key for recovering the original input.

TRANSFER IT TO AN EXAM ANSWER

Hash collisions are theoretically possible. A plain fast hash alone is not a recommended modern password-storage design.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.1(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain two distinct uses of hashing and why hashing is not the same as encryption.

QUESTION 2 | 4 MARKS

A file digest changes after downloading. Explain what can be concluded and what cannot be proved using only the digest.

1.3.1(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Integrity check: compare stored and calculated digests for changes.
- Password verification: compare a secure salted password-derived hash.
- Hashing is designed to be one-way, unlike reversible encryption.

Q2 / 4 MARKS

- Different digests indicate the file data differs or an error occurred.
- Matching values alone do not prove the sender is authentic.
- Hashes can collide; protect the trusted expected digest.

COMMON MISCONCEPTION

Do not describe a hash as “encrypted data that the user can decrypt with a key”.

Green pen: Name the threat being addressed: accidental change, tampering, or password disclosure.

1.3.2(a) / BUILD THE MENTAL MODEL

Link facts using stable identifiers

A relational database organises data into tables; keys describe how records are uniquely identified and related.

THREE PIECES OF THE IDEA

PRIMARY KEY

Uniquely identifies each row in its own table; values cannot be null.

FOREIGN KEY

A field or field set referencing a key in another table.

SECONDARY KEY

A non-primary field used for searching; it need not be unique.

MODEL / FOLLOW THE FLOW

Customer 1 → **Many Orders** → **Many OrderLines** → **Product reference**

Customer(CustomerID PK) links to Order(CustomerID FK): one customer can have many orders. A line table bridges an order to a product.

EXAM-READY STATEMENT

A foreign key is not necessarily unique. It is supposed to repeat in a one-to-many relationship.

WORDS THAT EARN MARKS

Entity Object of interest

PK Unique row identifier

FK Cross-table link

Cardinality 1:1, 1:M, M:M

APPENDIX EXAMPLES / OCR ERD cardinality >

1.3.2(a) / STEP-BY-STEP WORKED EXAMPLE

Turn a many-to-many relationship into two one-to-many links

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Recognise	A student takes many courses	A course also has many students: M:M.
2. Make entities	Student(StudentID, Name)	StudentID is the primary key.
3. Course	Course(CourseID, Title)	CourseID is the primary key.
4. Bridge	Enrolment(StudentID, CourseID)	Both foreign keys; composite primary key may be the pair.
5. Explain	Student 1:M Enrolment M:1 Course	The bridge prevents repeated course columns.

TRANSFER IT TO AN EXAM ANSWER

A foreign key is not necessarily unique. It is supposed to repeat in a one-to-many relationship.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Design three tables for customers placing orders. Label primary and foreign keys and explain the relationships.

QUESTION 2 | 4 MARKS

Explain the purpose of a secondary key and why an email address may be unsuitable as a primary key.

1.3.2(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Customer(CustomerID PK, Name).
- Order(OrderID PK, CustomerID FK, OrderDate).
- OrderLine(LineID PK, OrderID FK, ProductID FK, Qty) with a Product table if product details are stored.
- One Customer can have many Orders; one Order many OrderLines.

Q2 / 4 MARKS

- Secondary key improves locating/selecting rows on a searched attribute.
- Email can change, and uniqueness/availability may not be guaranteed.
- A stable surrogate identifier is often more suitable as the PK.

COMMON MISCONCEPTION

A many-to-many relationship requires an associative table in a conventional relational design.

Green pen: Draw the crow-foot direction and identify the table that holds each foreign key.

1.3.2(c) / BUILD THE MENTAL MODEL

Remove partial and transitive dependency

Normalisation restructures tables so each fact is stored in an appropriate place, reducing update, insertion and deletion anomalies.

THREE PIECES OF THE IDEA

1NF

Atomic cells and a key; no repeating groups or multi-valued fields in a cell.

2NF

1NF plus every non-key attribute depends on the whole candidate key, not just part of a composite key.

3NF

2NF plus no transitive dependency of non-key attributes on a key via another non-key attribute.

MODEL / FOLLOW THE FLOW

Flat sales list → **1NF line records** → **2NF split by key** → **3NF isolate customer**

Worked decomposition below uses OrderID + ProductID as the original composite key.

EXAM-READY STATEMENT

Exam explanations need the dependency that causes the problem, not only a list of resulting tables.

WORDS THAT EARN MARKS

- **1NF** Atomic fields
- **Partial** Part of key
- **Transitive** Non-key via non-key
- **Anomaly** Data inconsistency

1.3.2(c) / STEP-BY-STEP WORKED EXAMPLE

Normalise sales lines in stages

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Raw row	OrderID, ProductID, OrderDate, CustomerID, CustomerName, ProductName, Qty	One atomic product per line; key (OrderID, ProductID).
2. Partial	OrderDate relies only on OrderID	ProductName relies only on ProductID: violates 2NF.
3. Into 2NF	Order(OrderID, OrderDate, CustomerID, CustomerName); Product(ProductID, ProductName); Line(OrderID, ProductID, Qty)	Remove those partial dependencies.
4. Transitive	OrderID -> CustomerID -> CustomerName	CustomerName depends on CustomerID, not directly on OrderID.
5. Into 3NF	Customer(CustomerID, CustomerName); Order(OrderID, OrderDate, CustomerID); Product; Line	CustomerID is an FK in Order; line has two FKs.

TRANSFER IT TO AN EXAM ANSWER

Exam explanations need the dependency that causes the problem, not only a list of resulting tables.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

A table Sale(OrderID, ItemID, ItemName, Qty, CustomerID, CustomerName) uses key (OrderID, ItemID). Describe one partial and one transitive dependency and suggest a 3NF structure.

QUESTION 2 | 4 MARKS

Explain how 3NF can reduce an update anomaly when a customer changes their name.

1.3.2(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- ItemName depends on ItemID only: partial dependency.
- CustomerName depends on CustomerID, itself identified through OrderID: transitive dependency.
- Use Order(OrderID,CustomerID), Customer(CustomerID,CustomerName), Item(ItemID,ItemName), SaleLine(OrderID,ItemID,Qty), with appropriate PKs/FKs.

Q2 / 4 MARKS

- Store the customer name once in Customer.
- Order rows reference CustomerID rather than repeat the name.
- One update changes the authoritative value for all orders.

COMMON MISCONCEPTION

2NF and 3NF are about functional dependencies; simply saying “no duplicates” does not show the reasoning.

Green pen: Underline each determinant and dependent attribute before deciding which tables to create.

1.3.2(b) / BUILD THE MENTAL MODEL

Collect, validate, query, export

Data may enter a database through forms, sensors, OCR and file imports; it must be selected and exchanged appropriately.

THREE PIECES OF THE IDEA

CAPTURE

Choose collection method and validate data at input; validation does not guarantee truth.

SELECT

Filter records using searches, predicates, query-by-example or SQL.

EXCHANGE

Move data as files or over networks using agreed structures such as CSV or JSON.

MODEL / FOLLOW THE FLOW



A joined workflow: a parent submits a form, a database stores valid entries, and an authorised report exports selected records.

EXAM-READY STATEMENT

Validation confirms that input meets specified rules; verification compares data against the intended source.

WORDS THAT EARN MARKS

Validation Check rules

Verification Check accuracy

QBE Visual query design

CSV Delimited text

EXAM CHECK | OCR IS NOT OMR

OCR: recognises printed/scanned characters; e.g. text from a paper form.

OMR: detects marks in pre-defined positions; e.g. shaded MCQ answer boxes.

Neither proves the real-world data is correct; verify important entries.

1.3.2(b) / STEP-BY-STEP WORKED EXAMPLE

Safely import 120 student responses

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Capture	Form requires YearGroup as an integer	Use presence and range checks; collect only relevant data.
2. Verify	A valid postcode may still belong to the wrong person	Cross-check against a trusted source when appropriate.
3. Select	Return YearGroup = 12 and Consent = TRUE	Define a reproducible query filter.
4. Export	CSV: rows and fields; JSON: named structures	Choose format based on receiving system.
5. Protect	Control roles and sharing	Authorise access and transfer securely.

TRANSFER IT TO AN EXAM ANSWER

Validation confirms that input meets specified rules; verification compares data against the intended source.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

SECOND WORKED MODEL | PICK THE CAPTURE METHOD

Printed candidate number -> OCR; recognition errors may need checking.

Shaded option B bubble -> OMR; it detects a mark in a known position.

OMR does not translate freehand writing into text; OCR may misread print.

Validation tests rules; verification compares with the original source.

1.3.2(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Distinguish validation from verification in a school data-capture system, giving one example of each.

QUESTION 2 | 5 MARKS

Describe a suitable method to collect, select and exchange sensor data with a second application.

QUESTION 3 | 4 MARKS

A school scans shaded multiple-choice boxes and printed student names.
Choose OCR or OMR for each, and identify a realistic error for each.

1.3.2(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Validation: rule checks, e.g. a permitted YearGroup range.
- Verification: compare value with a known accurate source, e.g. confirm ID against original record.

Q2 / 5 MARKS

- Capture via sensor or interface into structured records.
- Validate required types/ranges and handle missing readings.
- Query records by timestamp or threshold.
- Export in a supported structured format such as CSV/JSON.
- Use protected transfer appropriate to confidentiality.

COMMON MISCONCEPTION

A field passing a range check does not prove that the field's information is correct.

Green pen: Give the rule and the resulting action when invalid data is detected.

Q3 / 4 MARKS | DATA CAPTURE

1 mark: OMR for positions of filled/shaded multiple-choice boxes.
1 mark: OCR for recognising printed student names as text.
1 mark: OMR may misread a faint/double mark or a misplaced shade.
1 mark: OCR may confuse similar characters; verify where required.

1.3.2(d) / BUILD THE MENTAL MODEL

Read the schema before writing the query

SQL retrieves related data by joining tables using matching keys and restricting the returned rows.

THREE PIECES OF THE IDEA

SELECT

Choose output fields. WHERE filters rows.

JOIN

Combine related tables through an explicit matching condition.

ORDER BY

Sort returned rows; use aggregation only when the task asks for summaries.

MODEL / FOLLOW THE FLOW

Student table → **StudentID match** → **Mark table** → **Filtered output**

For Student(StudentID,Name) and Mark(MarkID,StudentID,Score), join on Student.StudentID = Mark.StudentID.

EXAM-READY STATEMENT

SQL: SELECT Student.Name, Mark.Score FROM Student INNER JOIN Mark ON Student.StudentID = Mark.StudentID WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC;

WORDS THAT EARN MARKS

- **SELECT** Columns
- **FROM** Source table
- **JOIN** Combine rows
- **WHERE** Filter rows

APPENDIX EXAMPLES / Nested SELECT and SQL filters >

1.3.2(d) / STEP-BY-STEP WORKED EXAMPLE

Find learners scoring at least 70

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Schema	Student(StudentID, Name); Mark(MarkID, StudentID, Score)	StudentID in Mark is a foreign key.
2. Link	Student.StudentID = Mark.StudentID	Match each mark to its student.
3. Fields	SELECT Student.Name, Mark.Score	Return only the requested columns.
4. Tables	FROM Student INNER JOIN Mark ON ...	The join identifies matching rows.
5. Filter	WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC	Filter before presenting sorted output.

TRANSFER IT TO AN EXAM ANSWER

SQL: SELECT Student.Name, Mark.Score FROM Student INNER JOIN Mark ON Student.StudentID = Mark.StudentID WHERE Mark.Score >= 70 ORDER BY Mark.Score DESC;

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Write SQL to display student names and scores at least 80 using Student(StudentID,Name) and Result(StudentID,Score).

QUESTION 2 | 4 MARKS

Explain what would go wrong if a student/result join omitted the condition matching StudentID.

1.3.2(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- SELECT Student.Name, Result.Score
- FROM Student INNER JOIN Result
- ON Student.StudentID = Result.StudentID
- WHERE Result.Score >= 80;

Q2 / 4 MARKS

- Without an ON condition or correct join predicate, unrelated rows may be combined.
- This may produce a Cartesian product or otherwise incorrect results.
- Scores could be attributed to the wrong student.

COMMON MISCONCEPTION

Do not join using Name if an appropriate stable StudentID key exists.

Green pen: Before writing SQL, circle the output fields, underline the join keys, then write WHERE.

1.3.2(d,e) / BUILD THE MENTAL MODEL

Modify records without breaking the relationships

INSERT, UPDATE and DELETE change data. Referential integrity restricts foreign keys to valid referenced rows (or permitted nulls).

THREE PIECES OF THE IDEA

INSERT

Add a record; provide required values and respect constraints.

UPDATE

Change specified records; a missing WHERE may affect every row.

DELETE

Remove specified rows; foreign-key relationships may block deletion.

MODEL / FOLLOW THE FLOW

Valid customer PK → **Order customer FK** → **FK constraint** → **Safe update/delete**

A new Order(CustomerID=99) is rejected if CustomerID 99 does not exist and that foreign-key value must reference Customer.

EXAM-READY STATEMENT

Referential integrity prevents dangling references, but the precise response to parent deletion depends on the database's FK policy.

WORDS THAT EARN MARKS

- **Constraint** Data rule
- **Orphan** Broken FK link
- **UPDATE** Modify rows
- **DELETE** Remove rows

APPENDIX EXAMPLES / SQL INSERT / DELETE / DROP >

1.3.2(d,e) / STEP-BY-STEP WORKED EXAMPLE

Protect customer and order data

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Schema	Customer(CustomerID PK); Orders(OrderID PK, CustomerID FK)	Foreign key references Customer.
2. Insert	INSERT INTO Customer(CustomerID) VALUES (99);	Create parent row before child.
3. Child	INSERT INTO Orders(OrderID,CustomerID) VALUES (501,99);	Now the reference is valid.
4. Update	UPDATE Orders SET CustomerID=99 WHERE OrderID=501;	WHERE limits the change to one order.
5. Delete	DELETE FROM Customer WHERE CustomerID=99;	May be blocked while Order 501 references customer 99 unless cascading rule is configured.

TRANSFER IT TO AN EXAM ANSWER

Referential integrity prevents dangling references, but the precise response to parent deletion depends on the database's FK policy.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(d,e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 4 MARKS

Explain referential integrity using a Customer and Order table, including an invalid foreign-key example.

QUESTION 2 | 5 MARKS

Write an SQL UPDATE to set Quantity to 4 for OrderLine with OrderID=12 and ProductID=7. Explain why both keys matter.

1.3.2(d,e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 4 MARKS

- Order.CustomerID is a foreign key referring to Customer.CustomerID.
- A non-null reference such as 999 must match an existing Customer key.
- This avoids orphaned order records and inconsistent relationships.

Q2 / 5 MARKS

- UPDATE OrderLine
- SET Quantity = 4
- WHERE OrderID = 12 AND ProductID = 7;
- Both fields identify the intended row when their combination is the composite PK.

COMMON MISCONCEPTION

A foreign-key constraint does not mean every parent row must have a matching child.

Green pen: Say which row is the parent, which is the child and what the FK references.

1.3.2(f) / BUILD THE MENTAL MODEL

Both changes succeed, or neither does

A database transaction is a logical unit of work. ACID properties protect correctness when failures and concurrent users occur.

THREE PIECES OF THE IDEA

ATOMICITY

A transaction is all-or-nothing.

CONSISTENCY

Constraints and valid-state rules remain satisfied.

ISOLATION + DURABILITY

Concurrent transactions do not interfere incorrectly; committed results persist.

MODEL / FOLLOW THE FLOW

Begin transfer → **Debit account A** → **Credit account B** → **Commit or rollback**

One transfer must not debit money without corresponding credit, even if power fails mid-operation.

EXAM-READY STATEMENT

Locking can prevent another transaction changing a record until the first transaction completes, though it can reduce concurrency.

WORDS THAT EARN MARKS

- **Atomicity** All or nothing
- **Consistency** Rules hold
- **Isolation** Concurrent safety
- **Durability** Committed persists

1.3.2(f) / STEP-BY-STEP WORKED EXAMPLE

Transfer 20 between two account records

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Begin	BEGIN TRANSACTION;	Group operations as one unit.
2. Debit	UPDATE Account SET Balance=Balance-20 WHERE ID=1;	Account 1 is reduced.
3. Credit	UPDATE Account SET Balance=Balance+20 WHERE ID=2;	Account 2 gains 20.
4. Check	If any rule/check fails: ROLLBACK;	Atomicity restores initial consistent state.
5. Commit	Otherwise COMMIT;	Durability means committed changes survive a crash; locks/isolation avoid harmful interference.

TRANSFER IT TO AN EXAM ANSWER

Locking can prevent another transaction changing a record until the first transaction completes, though it can reduce concurrency.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.2(f) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 6 MARKS

Explain each ACID property for a bank transfer that moves 30 between two accounts.

QUESTION 2 | 5 MARKS

Two staff members update the last ticket at the same time. Explain how record locking can protect consistency and one disadvantage.

1.3.2(f) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 6 MARKS

- Atomicity: debit and credit both happen or both roll back.
- Consistency: account constraints remain true before/after.
- Isolation: simultaneous transfers do not see inconsistent partial states.
- Durability: committed balances persist after restart.

Q2 / 5 MARKS

- Lock the seat/inventory record while a transaction reserves it.
- Second transaction waits or receives a conflict.
- First commits before the other proceeds.
- Disadvantage: waiting, reduced throughput or possible deadlock.

COMMON MISCONCEPTION

ACID Consistency is about preserved integrity constraints, not a guarantee that every field value is factually true.

Green pen: For a failed transfer, explicitly state which operations are undone and why.

1.3.3(a,b) / BUILD THE MENTAL MODEL

Each layer handles a different responsibility

Protocol layering separates application services, end-to-end transport, addressing and local delivery.

THREE PIECES OF THE IDEA

APPLICATION

HTTP, DNS and other protocols used by application processes.

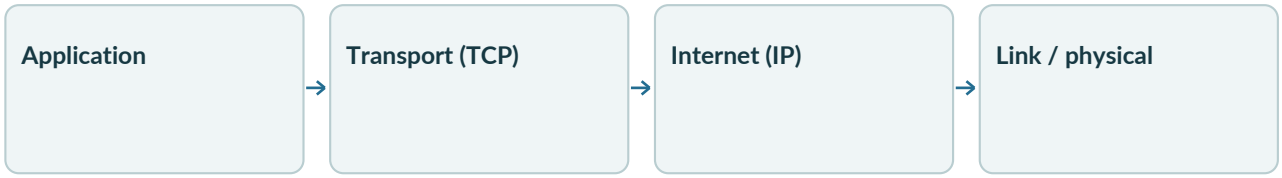
TRANSPORT + INTERNET

TCP provides ordered reliable streams; IP addresses and routes packets.

LINK

Local network frames and physical access such as Ethernet or Wi-Fi.

MODEL / FOLLOW THE FLOW



A four-layer TCP/IP teaching model shows encapsulation; some sources combine or split the bottom layers differently.

EXAM-READY STATEMENT

Protocol layering allows technologies to change at one layer while preserving well-defined interfaces to the others.

WORDS THAT EARN MARKS

- **Protocol** Agreed rules
- **Standard** Common specification
- **Layer** Responsibility
- **Packet** Network data unit

1.3.3(a,b) / STEP-BY-STEP WORKED EXAMPLE

Send a webpage request through the stack

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Application	Browser creates HTTP request	Application determines meaning of the message.
2. Transport	TCP segments stream and tracks delivery	Sequence/acknowledgement permits ordered reliable delivery in this example.
3. Internet	IP adds source/destination addressing	Routers forward towards destination network.
4. Link	NIC wraps packet for local network	Ethernet or Wi-Fi sends bits on next hop.
5. Receiver	Headers processed in reverse order	Application receives usable message content.

TRANSFER IT TO AN EXAM ANSWER

Protocol layering allows technologies to change at one layer while preserving well-defined interfaces to the others.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(a,b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Describe the roles of the application, transport, internet and link layers when a browser requests a webpage.

QUESTION 2 | 4 MARKS

Explain two benefits of protocols and common standards for devices made by different manufacturers.

1.3.3(a,b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Application: HTTP request format/semantics.
- Transport: TCP sequencing, acknowledgements and reliable delivery.
- Internet: IP addressing and route forwarding.
- Link: frames and local media delivery.

Q2 / 4 MARKS

- Devices use the same message formats and conventions.
- Interoperability avoids vendor-specific incompatible formats.
- Layers/components can be improved independently if standard interfaces are retained.

COMMON MISCONCEPTION

The protocol stack is not a literal list of four separate cables. Each layer supplies a service to the layer above.

Green pen: State one concrete responsibility and example per layer.

1.3.3(b) / BUILD THE MENTAL MODEL

The name is not the IP address

DNS resolves a domain name to DNS records including IP addresses. Routers use network addresses to forward packets.

THREE PIECES OF THE IDEA

DOMAIN

Human-readable name, for example school.example.

DNS

Distributed naming service returns records; local caches may speed repeated lookups.

IP + ROUTER

IP address identifies endpoints/interfaces; routers direct packets between networks.

MODEL / FOLLOW THE FLOW

Name entered → **DNS lookup** → **IP returned** → **Packets routed**

DNS maps names to records, not necessarily a permanent one-to-one server. A site may use multiple changing addresses.

EXAM-READY STATEMENT

DNS resolves names; a router forwards IP packets. Neither task alone guarantees encryption of the data being sent.

WORDS THAT EARN MARKS

- **DNS** Name resolution
- **IP** Network address
- **Router** Forwards packets
- **TTL** Cache lifetime

1.3.3(b) / STEP-BY-STEP WORKED EXAMPLE

Follow a request for a named service

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Name	Browser requests library.example	The URL host is a domain name.
2. Cache	Check cached DNS records	A valid cached answer may avoid a new lookup.
3. Resolve	DNS service returns address record	Result can be one of several possible IP addresses.
4. Connect	Client sends IP packets	Source/destination IPs support routing.
5. Route	Routers select next hops	Each router forwards based on its routing information.

TRANSFER IT TO AN EXAM ANSWER

DNS resolves names; a router forwards IP packets. Neither task alone guarantees encryption of the data being sent.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain what happens between entering a domain name and sending packets to a remote web server.

QUESTION 2 | 4 MARKS

Explain why DNS caching is useful and one reason a cached record might need refreshing.

1.3.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Client identifies hostname in the URL.
- Cached DNS answer may be used, otherwise resolver performs lookup.
- An address record is returned.
- Client sends IP packets addressed to the server.
- Routers forward packets towards destination.

Q2 / 4 MARKS

- Cache reduces lookup delay and query traffic.
- Records can change when services move or are load-balanced.
- TTL/expiry encourages refreshing stale values.

COMMON MISCONCEPTION

A hostname can map to several IP addresses; do not claim every website has exactly one fixed IP.

Green pen: Separate the naming question (DNS) from the delivery question (IP and routers).

1.3.3(b) / BUILD THE MENTAL MODEL

Shared routes versus reserved capacity

Packet switching divides a message into independently forwarded packets. Circuit switching reserves a route or capacity for a session.

THREE PIECES OF THE IDEA

PACKET

Shared network capacity; packets may take different paths and arrive out of order.

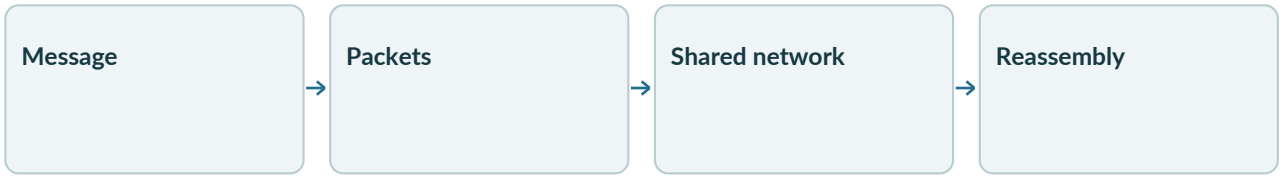
CIRCUIT

A dedicated path or reserved resources for the duration of communication.

CHOOSE

Bursty data favours efficient packet sharing; predictable reserved capacity may suit continuous real-time channels.

MODEL / FOLLOW THE FLOW



A file is split into packet payloads, routed over the internet, and reconstructed using higher-layer mechanisms.

EXAM-READY STATEMENT

Packet switching alone does not guarantee arrival in order; protocol mechanisms can provide that service.

WORDS THAT EARN MARKS

Packet	Routed unit	**Header**	Control details
Circuit	Reserved path	**Latency**	Transfer delay

1.3.3(b) / STEP-BY-STEP WORKED EXAMPLE

Send a 3 KB message as three 1 KB payloads

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Divide	P1, P2, P3 carry the data	Each packet gets addressing/metadata overhead.
2. Route	P1 via A; P2 via B; P3 via A	Shared network routes may differ.
3. Arrive	P2, P1, P3	Higher layers can reorder when required; packets may be lost.
4. Recover	TCP retransmits missing data in its stream	Reliability depends on protocols, not packet switching alone.
5. Compare	Reserved circuit stays allocated	Continuous predictable service but unused capacity can be wasted.

TRANSFER IT TO AN EXAM ANSWER

Packet switching alone does not guarantee arrival in order; protocol mechanisms can provide that service.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(b) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain why packet switching is appropriate for browsing but may suffer variable delay.

QUESTION 2 | 4 MARKS

Compare the benefits and disadvantages of packet switching and circuit switching for a fixed voice service.

1.3.3(b) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Traffic is bursty so many users share capacity.
- Packets are forwarded independently and can use available routes.
- Congestion, queueing and different paths introduce variable latency/jitter.
- Higher-layer protocols can provide reliability when needed.

Q2 / 4 MARKS

- Circuit reserves resources so delay/capacity can be predictable.
- Reserved capacity can be wasted during silence.
- Packet use is efficient but contention can cause delay/jitter.

COMMON MISCONCEPTION

Do not claim a circuit is established for ordinary IP packet switching.

Green pen: Link the network choice to occupancy, reliability and predictability requirements.

1.3.3(c) / BUILD THE MENTAL MODEL

One control cannot stop every threat

Security defences should match threats such as eavesdropping, malware, phishing, denial of service and SQL injection.

THREE PIECES OF THE IDEA

FIREWALL

Apply traffic rules; can block unwanted inbound or outbound connections.

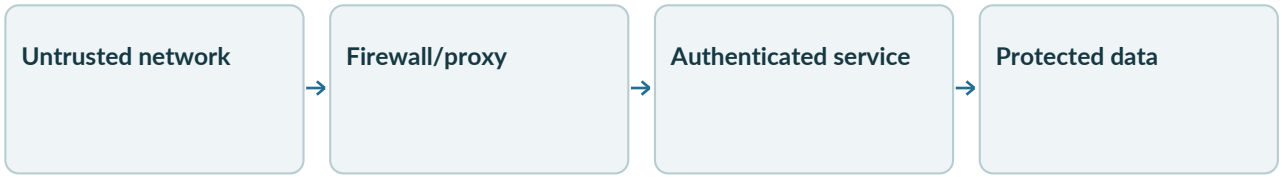
PROXY

Intermediary can filter requests, hide internal systems or cache content.

ENCRYPTION

Protects data confidentiality in transit when configured correctly; does not prevent all attacks.

MODEL / FOLLOW THE FLOW



Layered controls include user authentication, updates, backups and monitoring as well as perimeter devices.

EXAM-READY STATEMENT

A firewall filters traffic; it does not automatically remove malware already running or make application SQL safe.

WORDS THAT EARN MARKS

Phishing Deceptive message

DoS Disrupt service

TLS Secured transport

Injection Unsafe query input

1.3.3(c) / STEP-BY-STEP WORKED EXAMPLE

Protect a school results portal

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Threat	Attacker intercepts logins on public Wi-Fi	Use HTTPS/TLS so traffic is encrypted in transit.
2. Access	Attacker guesses a staff password	Require MFA and rate limiting; train staff.
3. Injection	User enters malicious SQL into a search box	Use parameterised queries and input validation.
4. Flood	Many requests exhaust server capacity	Rate limiting, filtering and DDoS protection may help.
5. Recover	Ransomware damages local files	Use tested backups and controlled permissions.

TRANSFER IT TO AN EXAM ANSWER

A firewall filters traffic; it does not automatically remove malware already running or make application SQL safe.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A results portal suffers interception attempts and SQL injection. Recommend separate controls and explain each.

QUESTION 2 | 5 MARKS

Compare a firewall, proxy and encryption: state their distinct purposes and one limitation of each.

1.3.3(c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Use TLS/HTTPS to protect data from passive interception.
- Use parameterised SQL to separate input values from query structure.
- Access controls/MFA further protect sensitive results.

Q2 / 5 MARKS

- Firewall controls traffic using rules but cannot stop all application attacks.
- Proxy intermediates/filters requests but is not inherently encryption.
- Encryption protects confidentiality but not availability or compromised endpoints.

COMMON MISCONCEPTION

Encryption does not protect against a legitimate user being fooled into entering credentials on a phishing website.

Green pen: Pair each named threat with a mechanism that blocks or reduces it.

1.3.3(d) / BUILD THE MENTAL MODEL

Choose the device by what it forwards

Devices operate at different levels: hosts need interfaces; local networks need switching; separate networks need routing.

THREE PIECES OF THE IDEA

NIC + SWITCH

NIC connects a host; a switch forwards frames within a LAN using MAC addresses.

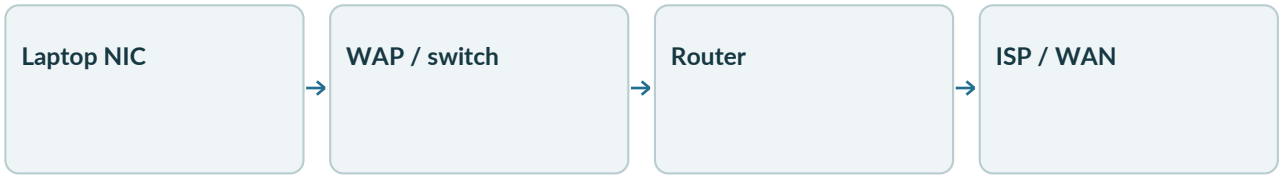
ROUTER

Forwards packets between IP networks using routing information.

WAP + MODEM

Wireless access point bridges Wi-Fi clients into a LAN; a modem adapts a link to an access technology.

MODEL / FOLLOW THE FLOW



A laptop joins Wi-Fi via an access point, local traffic uses switching, and a router sends traffic to remote networks.

EXAM-READY STATEMENT

A switch normally handles local Ethernet forwarding; a router connects distinct IP networks.

WORDS THAT EARN MARKS

- **NIC** Host interface
- **Switch** LAN frames
- **Router** Between networks
- **WAN** Wider network

1.3.3(d) / STEP-BY-STEP WORKED EXAMPLE

Design a school wired/wireless layout

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Devices	Desktop NICs connect using Ethernet	Wired endpoints plug into an access switch.
2. Wi-Fi	Mobile devices join the WAP	Wireless clients are bridged to the local network.
3. LAN	Switch sends local frames appropriately	MAC learning reduces unnecessary forwarding.
4. Exit	Router connects LAN to another IP network	Packets for outside subnets are forwarded.
5. WAN	ISP link reaches distant networks	The WAN connects geographically separated networks.

TRANSFER IT TO AN EXAM ANSWER

A switch normally handles local Ethernet forwarding; a router connects distinct IP networks.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

A school expands to two buildings with Wi-Fi and an internet connection. Name suitable network hardware and justify each.

QUESTION 2 | 4 MARKS

Explain the difference between LAN and WAN using ownership and geographic scope, with an example.

1.3.3(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- NICs provide host connections.
- Switches connect wired LAN endpoints.
- WAPs provide wireless access.
- Router forwards between school network and internet/WAN.
- Appropriate medium/links connect buildings.

Q2 / 4 MARKS

- LAN covers a local site/campus, often managed by one organisation.
- WAN spans larger geographic regions and connects LANs, often across provider links.
- Example: school LAN versus school-to-trust WAN.

COMMON MISCONCEPTION

A Wi-Fi access point is not necessarily the router, although home equipment often combines both functions.

Green pen: Name the required movement: host-to-LAN, LAN-to-LAN or LAN-to-WAN.

1.3.3(e) / BUILD THE MENTAL MODEL

Where does responsibility for a service sit?

Client-server networks rely on servers to provide services to requesting clients. Peer-to-peer networks let nodes provide services directly to one another.

THREE PIECES OF THE IDEA

CLIENT-SERVER

Centralised services, accounts, security policy and backup are easier to manage.

PEER-TO-PEER

Devices act as both service consumers and providers; may avoid a central server.

TRADE-OFF

Consider administration, resilience, expense, scalability and control of data.

MODEL / FOLLOW THE FLOW



In a school, a central file server controls access; on a simple peer-share network, computers share files directly.

EXAM-READY STATEMENT

Client-server is a logical relationship, not a rule that every network needs one physical central computer.

WORDS THAT EARN MARKS

Client	Requests service	**Server**	Provides service
Peer	Equal network node	**Centralise**	One control point

1.3.3(e) / STEP-BY-STEP WORKED EXAMPLE

Choose an architecture for two scenarios

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Exam files	100 users need audited access	Dedicated file server supports permissions and backup.
2. Updates	Teacher changes a shared document	Central server holds authoritative version.
3. Small group	Three laptops share temporary files	Peer-to-peer can work with little setup.
4. Availability	Server fails at exam time	Client-server needs reliable infrastructure/recovery.
5. Security	Peer machine is compromised	Distributed data sharing is harder to administer consistently.

TRANSFER IT TO AN EXAM ANSWER

Client-server is a logical relationship, not a rule that every network needs one physical central computer.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.3(e) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Recommend an architecture for a 500-user school needing controlled student file access and justify its limitations.

QUESTION 2 | 4 MARKS

Explain two differences between peer-to-peer and client-server networks with real implications.

1.3.3(e) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Use client-server for central authentication/access controls.
- Central backups, audit logs and consistent administration suit the scale.
- Costs include dedicated infrastructure and managing server failures.

Q2 / 4 MARKS

- Client-server requests go to a providing server; peer nodes both request and provide resources.
- Central management simplifies backups and access control.
- P2P can be simpler/cheaper but harder to secure consistently.

COMMON MISCONCEPTION

Peer-to-peer does not mean “there is no security”; access can be configured, but it is decentralised.

Green pen: Choose the architecture by citing user numbers, control needs and failure consequences.

1.3.4(a) / BUILD THE MENTAL MODEL

Structure, presentation, interaction

HTML defines document content and structure, CSS styles it and JavaScript can update content and respond to user input.

THREE PIECES OF THE IDEA

HTML

Elements such as headings, paragraphs, links and forms define the content structure.

CSS

Selectors and declarations control colour, spacing, layout and fonts.

JAVASCRIPT

Code can read inputs, respond to events and modify the DOM.

MODEL / FOLLOW THE FLOW

HTML element → **CSS selector** → **JS event** → **Updated webpage**

Concept: an HTML paragraph with id="msg" gets CSS style and is changed by a JavaScript event listener.

EXAM-READY STATEMENT

HTML does not execute the JavaScript just because a tag is present; the browser runs script code in the appropriate context.

WORDS THAT EARN MARKS

- **Tag** Markup syntax
- **Selector** CSS target
- **DOM** Page object tree
- **Event** User action

APPENDIX EXAMPLES / HTML tag and form practice >

1.3.4(a) / STEP-BY-STEP WORKED EXAMPLE

Interpret a small page with a button

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. HTML	Ready	Creates a paragraph with identifier msg.
2. HTML	<button id="go">Start</button>	Creates an interactive button.
3. CSS	#msg { color: navy; }	Changes the paragraph's visual presentation.
4. JS	document.getElementById("go").onclick = function(){...}	Registers behaviour to run on clicking.
5. JS body	document.getElementById("msg").textContent = "Go";	Changes text shown in the existing paragraph.

TRANSFER IT TO AN EXAM ANSWER

HTML does not execute the JavaScript just because a tag is present; the browser runs script code in the appropriate context.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(a) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Explain how HTML, CSS and JavaScript work together to display and update a paragraph after clicking a button.

QUESTION 2 | 5 MARKS

A page has 0 . Give CSS to make text blue and one JavaScript statement to set the text to 10.

APPENDIX EXAMPLES / CSS and JS appendix practice >

OCR H446/01 / STUDENT REVISION

1.3.4(a) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- HTML paragraph/button defines structure and identifiers.
- CSS selects element and changes appearance.
- JavaScript attaches event handler to button.
- When clicked, handler updates DOM text content.

Q2 / 5 MARKS

- CSS: #score { color: blue; }
- JS: document.getElementById("score").textContent = "10";
- Identifier spelling/case must match the HTML.

COMMON MISCONCEPTION

CSS changes presentation, not the underlying meaning of an HTML element; JavaScript is not the same as Java.

Green pen: Point to the element, style rule, event and output separately.

1.3.4(b,c) / BUILD THE MENTAL MODEL

Discover pages; index content; rank importance

A search engine uses crawlers to discover pages, builds an index and then orders results using ranking signals. PageRank measures link-based importance.

THREE PIECES OF THE IDEA

CRAWL

Automated crawler follows links and discovers pages.

INDEX

Extract terms and metadata into a searchable structure.

RANK

Rank candidates using relevance and link-based authority; PageRank treats inbound links as weighted votes.

MODEL / FOLLOW THE FLOW

Web crawler → **Content index** → **Matching pages** → **Ranking order**

Page A receiving a link from an important page can earn more link importance than a link from an unimportant page.

EXAM-READY STATEMENT

PageRank is about the quantity and quality of inbound links, not a count of how many times a keyword appears.

WORDS THAT EARN MARKS

- **Crawler** Discovers pages
- **Index** Searchable entries
- **Inbound** Link pointing in
- **Damping** Random-jump effect

1.3.4(b,c) / STEP-BY-STEP WORKED EXAMPLE

Compare links in a tiny web graph

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Pages	A, B and C are web pages	Create directed arrows for hyperlinks.
2. Links	B -> A; C -> A; A -> B	A receives two inbound links; B receives one.
3. Quality	C is itself authoritative	A link from C may pass more weight than a weak page.
4. Share	A divides its outgoing rank across links	The share depends on the number of outgoing links.
5. Damping	Allow a small chance to jump to any page	Reduces traps/dead ends and supports stable scores.

TRANSFER IT TO AN EXAM ANSWER

PageRank is about the quantity and quality of inbound links, not a count of how many times a keyword appears.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(b,c) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Outline how a search engine discovers, indexes and ranks a new page, including PageRank.

QUESTION 2 | 4 MARKS

Page X has ten inbound links from low-quality pages; Y has one from an authoritative page. Explain why Y may rank higher on PageRank.

1.3.4(b,c) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Crawler follows links/discovers page.
- Indexer stores processed content and searchable terms.
- Query retrieves candidate documents.
- Ranking combines evidence; PageRank uses the importance of linking pages.

Q2 / 4 MARKS

- Inbound links act as weighted endorsements.
- Importance passed by a high-rank page can outweigh several weak links.
- Rank is also split across the linking page's outgoing links.

COMMON MISCONCEPTION

PageRank is not the same as the entire search-engine ranking algorithm; relevance and other signals also matter.

Green pen: When explaining a ranking, trace the direction of each hyperlink arrow.

1.3.4(d) / BUILD THE MENTAL MODEL

Run it close to the user or beside the data?

Client-side code executes in the browser; server-side code executes on the server. Most interactive sites use both.

THREE PIECES OF THE IDEA

CLIENT

Fast local feedback, page interaction and initial field validation.

SERVER

Handles trusted rules, database queries, sessions and private data.

SECURITY

Never trust browser validation alone: it can be bypassed. Recheck on the server.

MODEL / FOLLOW THE FLOW

Browser enters → **Client validates** → **Server verifies** → **Database query**

Shopping checkout: JS flags an empty postcode locally; server validates price/stock before writing an order.

EXAM-READY STATEMENT

Client-side checks are for convenience; essential security, permissions and consistency must be enforced server-side.

WORDS THAT EARN MARKS

Client Browser device

Server Remote service

Validation Rule check

Trust boundary Untrusted input

APPENDIX EXAMPLES / JavaScript program tracing >

1.3.4(d) / STEP-BY-STEP WORKED EXAMPLE

Process an online order safely

Follow each stage in order. Explain both the change and why it happens before you try the exam questions.

TRACE / DECISION TABLE

STEP	VALUE OR ACTION	REASON / INTERPRETATION
1. Form	Customer enters delivery address	HTML fields collect user input.
2. Client JS	Show missing postcode immediately	Reduces waiting; improves usability.
3. Submit	Browser sends request to server	Client data must be treated as untrusted.
4. Server	Check identity, stock and price	Authoritative rules execute at protected server.
5. Store	Server transaction saves the order	Database operation happens outside the browser.

TRANSFER IT TO AN EXAM ANSWER

Client-side checks are for convenience; essential security, permissions and consistency must be enforced server-side.

Exam habit: state the condition, describe the mechanism, and link it to the specific outcome. Check intermediate values and key relationships, not just the final result.

1.3.4(d) / INDEPENDENT EXAM PRACTICE

Use the model without copying it

Original teacher-written practice questions (not OCR past-paper questions). Work in black pen; use the next page to check.

BLACK PEN: ANSWER FIRST. THEN CHECK THE SEPARATE MARK POINTS.

QUESTION 1 | 5 MARKS

Compare client-side and server-side processing for an online shop, giving two tasks for each and a limitation.

QUESTION 2 | 5 MARKS

Explain why a JavaScript check that a price field is positive is not enough to secure an online purchase.

1.3.4(d) / SELF-MARKING / INDICATIVE ANSWERS

Mark, correct, explain

Check method and vocabulary, not just the final number. These are teacher-written points, not an OCR mark scheme.

Q1 / 5 MARKS

- Client JS can validate form format and update page instantly.
- Server checks stock, authorisation and writes database changes.
- Client can be manipulated or disabled; server has processing/network costs.

Q2 / 5 MARKS

- Client-side code/input can be modified or bypassed.
- An attacker can send a crafted request without using the original form.
- Server must validate price/permissions and perform authoritative calculations.
- Transaction checks and safe database access protect integrity.

COMMON MISCONCEPTION

JavaScript can run on a server too; “JavaScript” alone does not prove execution location. Say where the code runs.

Green pen: Label each stage browser or server and name the reason for that choice.